

WLAN PROS TOOLBOX 1.11

Wireless Classroom: Teacher's Guide

How to present it, what each tool teaches, and lesson sequences that work

Keith Parsons, WLAN Pros · September 2026

Wireless Classroom • Teacher's Guide

The Wireless Classroom is the part of the WLAN Pros Toolbox built for teaching. It holds interactive simulators, 29 guided lessons, and the course handouts I use in class. It ships in the same free app your students already have, so whatever you show on the projector, they can open on their own laptop or tablet after class and work through at their own pace.

This guide covers how to present it, what each tool teaches, and a few lesson sequences that work well.

Before class

Use a computer or a tablet. The simulators are designed for a large screen. On a phone they open behind a short notice that says so, with a Continue anyway button. The guided lessons and the handouts work on any screen.

No Internet needed during class. The simulators compute everything on the device, and the lessons and handouts are built into the app, so a room with poor Wi-Fi does not stop the lesson. Install or update the app before class, and open the Wireless Classroom once to check it is all there.

Set the scene first. Open a tool, press the 'Present' button on the top right to have it go full screen. Then set it up the way you want to start (a wall material, a channel plan, a number of stations). I've found it easier to set the various options while already in Presenter mode.

Presenting

Every simulator has a **Present** button in its title bar. It appears whenever the window is at least 720 by 480 pixels, so you will see it on a laptop or tablet but not on a phone.

Present gives you one screen with no scrolling: the animation fills about two-thirds of the width, the controls sit in a panel on the right, and the most important number moves onto the stage in large type so the back of the room can read it. On a Mac it goes full screen. Text and lines scale up with the window, because a projector makes any window the same size on the wall.

The keys work the same way in every simulator:

Key	What it does
Space	Play or pause
Right arrow	Step forward one step (a slot, a frame, a message, a segment)
R	Reset
Up and Down	Move the tool's main control (distance, SNR, channel width, and so on)
F	Switch full screen on or off
?	Show every key for this tool
Esc	Leave presenter mode

Some tools add their own keys, and ? always lists them:

Tool	Its own keys
Fourier and FFT	1 to 4 switch modes; W changes the FFT window
DFS and Radar	D fires a radar event
802.1X and EAP Ladder; Association, Frame by Frame	Left takes a message back
Multi-Link Operation	N draws new traffic
Uplink vs Downlink	P reveals the answer; M turns the AP down to match, and back
Body Loss	Left and Right turn the holder; P reveals the answer
Channel Utilization Meter	B adds a one-second burst; W skips ahead one window
Heat Map Builder	E switches heat map, truth and error; W shows the hidden wall; P steps the power; N is the next step
Predict, Then Measure	Space reveals the truth; M steps through the maps; B walks both sides of every wall; O walks one side
Where Am I?	M (or Tab) switches the method; B blocks one AP's direct path; P steps the -70 dBm question

Tools with nothing to play or step ignore Space and the Right arrow.

The bar at the top (title, a light and dark switch, and Exit) hides itself when the mouse is still and comes back when you move it. If the projector washes out the dark theme, switch to light mode.

A teaching pattern that works: predict, then reveal

Before you press a key, ask the room what will happen. "If I move this client twice as far from the AP, how much signal do we lose?" "If one client drops to 6 Mbps, what happens to everyone else?" Take the guesses, then press the key. The simulators are built so one control makes one visible change, which is exactly what you need for this. Students remember the moment they were wrong far longer than a slide that told them the answer.

What each shelf teaches

Guided Lessons

- **Antenna Fundamentals.** A read-along lesson on what an antenna does (it shapes where the energy goes; it does not add power), gain against beamwidth, polarization, downtilt, and how to read a radiation pattern. Pair it with Antenna Pattern.
- **Spectrum Analysis.** A read-along lesson on why a spectrum analyzer sees energy a Wi-Fi adapter cannot, how the instrument works, and the signatures of common interferers. Pair it with the Swept vs FFT race in Fourier and FFT.
- **Find My, Explained.** How Apple finds your people, your phone and your things, including the AirTag in your suitcase, and what you should turn on before your next trip.
- **Connected, No Internet: Captive Portals.** Why a hotel or airplane network shows full bars and still won't load anything. Step through the device's check after association and flip the switch between a network that announces its portal and one that intercepts traffic. Stop at step 3 and ask the room what the device will see. It also explains why Wi-Fi Calling waits for the sign-in. Pair it with Association, Frame by Frame, which plays every frame up to DHCP and DNS.
- **Why the TV and the Printer Vanish on Guest Wi-Fi.** Why screen casting and printing find nothing from a guest network. Pick Same, Guest or Isolation and step the discovery question to the point

where it stops. Ask the room first where they think it will stop with isolation on. Pair it with Multicast at the Basic Rate, which covers what multicast costs in airtime.

- **Starlink, Explained.** How internet from space reaches your house or RV, how high the satellites are, and why the Wi-Fi inside matters as much as the sky above.
- **Wi-Fi Calling, Explained.** What happens when your phone makes a call over Wi-Fi instead of the cell tower, when it helps, when it drops, what it means for emergency calls, and what to set up before your next trip.
- **Home Internet, Explained.** Fiber, cable, DSL, 5G and satellite: how each one reaches a house, what to check before you sign up, and why the Wi-Fi is a separate question.
- **Phone Data Abroad, Explained.** Roaming with your own carrier, a travel eSIM and a local SIM, how to keep your own number on while a second plan carries the data, and the switches that stop a surprise bill.
- **Smart Home Radios, Explained.** Wi-Fi, Thread, Zigbee and Matter: three radios and a language, why a 2.4 GHz-only smart plug won't join, the gadget network that fixes it, and what a Thread border router does. Figure 3 (one band, shared) is the Wi-Fi hook for a Wi-Fi audience.
- **Cameras, Doorbells and Baby Monitors, Explained.** A camera is the rare gadget that sends more than it receives: upload against download, signal through the outside wall, what a wired doorbell's wire carries, cloud or home recording, baby monitors, locking a camera down, and jammers. Figure 3 (same house, same doorbell) is the Wi-Fi hook for a Wi-Fi audience.
- **How Your Devices Access the Internet, Explained.** Wi-Fi and cellular are ways to reach the internet, not the internet: two roads that meet at one box, six stops, what the bars measure, and why a bigger plan won't fix slow Wi-Fi. Figure 9 (the hotel with perfect bars) is the story to tell a room. Pair it with Test My Connection, which grades the two roads separately.
- **Mesh, Extenders and Wired Access Points, Explained.** Three ways to fix a dead zone, and the one question that decides between them: how does the new box talk to the main one? One radio taking turns, where to put an extender, five setups for one far room, and four ways to build the link back. Pair it with Repeaters and Mesh Backhaul, the model its numbers come from.
- **Is Your Router Too Old?** A router is too old when it stops getting security fixes, not when it stops working: the three stages, the five-minute check, who updates the box, keep, update or replace, the 2026 US router rule, and what to look for in the next one.

- **Tap, Touch and Near, Explained.** The two short-range radios in a phone: NFC, the one that works when you tap to pay, and Ultra Wideband, the one that measures how far away something is. Why a tap has to be a tap.
- **Satellite Texting, Explained.** How a phone sends a text through a satellite when there is no cell signal and no Wi-Fi, why it needs a clear view of the sky, and why it is a backup and not a promise.
- **Wi-Fi on Planes, Explained.** What airplane mode turns off, how the internet reaches a plane, why it feels slow when everyone shares one link, and what to do when the sign-in page won't appear.
- **Wi-Fi and Your Health, Explained.** How much power a router sends, how fast the signal fades, how far below the safety limits it measures in real homes and schools, and how the limits are set.
- **Travel Routers, Explained.** The small box that joins the hotel Wi-Fi for you and makes your own Wi-Fi in the room: its two jobs, the hotel's device limit, and setting it up at home. Pair it with Captive Portals, Explained.
- **Mobile Hotspots, Explained.** Turning the cell signal into Wi-Fi, with a pocket box or your phone: what decides the speed, data and your plan, battery and heat, and where to put it.
- **Captive Portals, Explained.** The Wi-Fi sign-in page from the user's side: how the phone notices it, what to do when the page doesn't show, why it comes back, and how to stay safe on it. Connected, No Internet covers the network side.
- **Bluetooth, Explained.** Why earbuds cut out and the car won't connect, and the three usual causes: something in the way, a crowded band, or a pairing that has gone stale.
- **Weak Cell Signal at Home, Explained.** Why the phone has bars at the curb and drops calls in the kitchen. Usually it is the building. It compares the three fixes: Wi-Fi Calling, a signal booster and a network extender.
- **How GPS Works, Explained.** How a phone finds itself by listening to satellites that send the time and where they are, and what to do when the blue dot gets it wrong.
- **How Your Phone Knows Where It Is, Explained.** The five kinds of clues a phone blends into one best guess, indoors and out, and which switches are yours.
- **Analog vs Digital, Explained.** What the two words mean, and why a Wi-Fi link slows down while a TV picture just freezes.

- **Scales and Ratios, Explained.** The tiny and huge numbers behind Wi-Fi turned into things you can hold: a foot of wire for a nanosecond, a stack of money for decibels, a ruler for a radio wave, jars for bytes.
- **Public Wi-Fi.** What the person next to you on public Wi-Fi can see. Pick Open, Enhanced Open or Password and watch which stretch of the trip each seal covers.
- **Wi-Fi Privacy Myths.** Your phone's private Wi-Fi address (Off, Fixed or Rotating) and what two routers and one cafe record for the same phone, then the hidden network name.

RF and Propagation

- **FSPL Simulator.** Free-space path loss against distance for 2.4, 5 and 6 GHz on one chart, with a Why panel that splits each band's loss into the part every band shares and the part that changes with frequency. Up and Down double and halve the distance, so each press shows the 6 dB per doubling rule. Log and Linear switch the distance axis: on log the bands are straight lines, on linear they are the curve students usually expect.
- **Wi-Fi Through a Wall.** One wave meets one wall: part reflects, the wave gets smaller as the material absorbs it, and the same wave continues behind the wall, smaller. The frequency never changes. Pick a real wall, such as an interior stud wall (plasterboard, air gap, plasterboard) or a concrete elevator-shaft wall, or one material at any thickness. Up and Down step through the walls; minus and equals make One material thinner and thicker.
- **Multipath Simulator.** Why the signal changes when you move a few centimeters: the direct signal and its reflections arrive with different phases and add like arrows. Up and Down move the receiver a sixteenth of a wavelength, so a few presses walk from a peak into a null.
- **Room Propagation.** An AP, walls and doorways on a floor plan, colored by received power, with the loss at a spot split into free space, walls and diffraction for each band.
- **Antenna Pattern.** A radiation pattern in 3D beside the two cuts a datasheet prints. Space spins it. Turn up the gain on an omni and watch it flatten, then flip between ceiling and wall mounting.
- **Rate vs Range.** One ring per MCS around an AP, and a cell edge where beacons sent at the minimum basic rate can no longer be decoded. Up and Down move the client out and in.
- **6 GHz Power and PSD.** The most power each 6 GHz device class may radiate, and the SNR it leaves, against channel width. Up and Down step the width, which shows when a wider channel keeps its SNR and when it loses 3 dB per doubling.
- **Why Two Devices Disagree.** Two devices at the same spot, hearing the same AP, report different signal strength. The true power is held fixed and each device reports it its own way, with an offset, grip and orientation loss. Ask the room which one is right before you click Reveal. Space re-samples; Up and Down move the AP 1 m (1 ft when lengths are imperial).

- **Uplink vs Downlink.** Both directions of one link at once: the two ends usually transmit at different power, so each can hear the other over a different distance, and the shaded band between the two rings is where the link is lopsided. Up and Down move the client; M turns the AP down to match and back, which answers "should I just turn the AP down?"
- **Body Loss.** An auditorium, one AP, a person holding a device and up to 50 other people. Turn the holder's back to the AP and watch their own body take the signal; fill the room and watch the crowd take more. Left and Right turn the holder; Space empties or fills the room.

Signals and PHY

- **Modulation Simulator.** How bits become a radio wave: each group of bits picks a point on the constellation, and noise pushes the received point toward its neighbors. Up and Down move the SNR; watch the errors appear.
- **Fourier and FFT.** Four modes: build a signal from sines, see how an FFT analyzer measures it, race a swept analyzer against an FFT analyzer on Bluetooth and a microwave oven, and see OFDM as an inverse FFT. Keys 1 to 4 switch modes.
- **OFDMA Resource Units.** One channel split into resource units so an AP can serve several clients in one transmission, compared with sending to each in turn.
- **MIMO and Beamforming.** Why a 4x4 AP and a 2x2 client use two streams, what the spare antennas do instead, what beamforming costs, and why a capture taken nearby misses beamformed frames. Up and Down steer the client.
- **PHY Preamble Reference.** The preamble of every OFDM PPDU format from 802.11a to Wi-Fi 7 (802.11be) drawn to scale, with each SIG field's bits. The Which PHY? mode walks the room through how a receiver tells the formats apart.

Airtime and Access

- **Medium Access Simulator.** Stations sharing one channel, slot by slot: backoff, collisions, the contention window doubling, EDCA priority, hidden nodes and RTS/CTS.
- **Voice Priority, End to End.** Follow one voice packet to a phone on Wi-Fi and see which of the AP's four queues it waits in. The call gets the Voice queue only if its EF (Expedited Forwarding) marking survives every hop and the AP maps it to the right priority; lose it at the tunnel, the provider or the AP, and the call waits in Best effort behind the download. Up and Down move where the marking is lost; D starts or stops the download.
- **Airtime Anatomy.** One transmit opportunity drawn to scale, microsecond by microsecond. The Right arrow walks it one segment at a time. Compare one frame against 32 aggregated frames and ask how much of the air carried data.

- **Airtime Fairness.** Why one slow client drags every fast client down, and how airtime fairness changes that. Up and Down switch the sharing rule.
- **Rate Adaptation.** A link learning its best rate frame by frame, the way Linux rate control does. Walk the client away and back and watch the rate step down and recover.
- **What an Interferer Costs.** How a Wi-Fi radio decides the air is busy: it holds off at -82 dBm for another Wi-Fi transmitter it can decode (preamble detect), and only at -62 dBm for anything else (energy detect). That is why a neighbor's AP on your channel costs more airtime than a microwave oven at the same level. Up and Down move the selected source's level; N picks the next source.
- **Spatial Reuse.** Two networks on one channel, and what BSS coloring and OBSS_PD let an AP do. Up and Down move the threshold, and the links show what each step costs.
- **Power Save.** Beacons, DTIM, and four ways a client sleeps, with awake time, battery and latency for each. Up and Down change the DTIM period.
- **Multi-Link Operation.** Wi-Fi 7 MLO modes compared on the same traffic, including cases where using more links is worse. N draws new traffic.
- **Multicast at the Basic Rate.** Why a multicast stream can eat the channel: by default it goes at a basic rate with no acknowledgment and no retry (Groupcast with Retries and multicast-to-unicast conversion are the exceptions), so the same bytes take many times the airtime of unicast. Compare the stream with the same stream converted to unicast. Up and Down change the basic rate.
- **Channel Utilization Meter.** The number an AP puts in its beacon as channel utilization, how it is averaged, and what it does not tell you. Space runs the channel; Up and Down change the number of senders; B adds a one-second burst.
- **Legacy Protection Cost.** What one associated 802.11b device costs a 2.4 GHz network, even while it sends nothing: a slow protection frame in front of every modern frame, and a longer slot time. At the slowest legal protection rate, a 54 Mb/s laptop's ceiling roughly halves. The beacon inspector shows the protection bits switch on.

Network Design and Security

- **Channel Planner.** Access points on a floor with channels and widths, showing which ones share airtime. Up and Down change the channel width for every AP at once; try the Auto-plan.
- **Roaming Walk.** A client walks a floor of APs, and the screen shows when it roams and what each roam costs. Up and Down move the roam trigger, so you can make a sticky client and a client that bounces between APs.
- **DFS and Radar.** One AP on a simulated one-hour clock: the channel availability check, a radar event (press D), the move, and the 30-minute lockout.
- **802.1X and EAP Ladder.** An 802.1X connection one message at a time across the client, the AP and the RADIUS server, for EAP-TLS, PEAP and EAP-TTLS. WPA2-Personal (PSK) and WPA3-Personal

(SAE) are there for contrast: no RADIUS server and no EAP conversation. Right sends the next message; Left takes one back. Roam mode plays a roam instead of a first connection, so you can compare a full 802.1X reconnect with a fast roam.

- **Association, Frame by Frame.** Every frame a client sends and receives to associate with a network, from the first scan to the first useful packet: authentication, association, the 4-way handshake, DHCP, then ARP and DNS, with a timeline of how long each phase takes. Right sends the next frame; Left takes one back.
- **Adjacent Channels and AP Stacking.** Why a channel that does not overlap yours still hurts when the other radio is close: its transmit mask leaks into your channel, and distance is what saves you. Start with the question of two APs 30 cm apart. Up and Down move the neighbor.
- **Band Steering.** Why a dual-band client so often stays on 2.4 GHz, and what an AP can and cannot do about it. The client chooses; the AP can only hide, refuse, suggest or deauthenticate it (disconnect it), and none of those stops the 2.4 GHz beacons. A deauthenticated client picks its band again, so it may come straight back to 2.4 GHz and be deauthenticated again, with no traffic each time. Space walks the client; Up and Down move it 5 m.
- **Survey Walk.** Why walking speed matters in a site survey: a scanner visits one channel at a time, so each channel's samples land far apart when you walk fast. White on the map means no data, never no coverage. Up and Down change the walking pace.
- **Heat Map Builder.** Build a heat map from samples you place, and see which cells are measurements and which are guesses. Change the interpolation, average in dB or in milliwatts, and watch the map change. Up and Down move the guess range; E switches between the heat map, the truth and the error.
- **Predict, Then Measure.** Test a predictive design against the building with an AP on a stick. Every wall loss in a prediction is a claim; walk the floor and see which walls the design got wrong, and which your walk never tested. Space reveals the truth; B walks both sides of every wall.
- **Where Am I?** Find a device two ways, by signal strength and by round-trip timing (FTM), and see how far off each one is. B blocks one AP's direct path, which throws both estimates off in different ways.
- **Repeaters and Mesh Backhaul.** What relaying costs: each hop's rate and throughput, and why a same-channel repeater makes every frame cross the air twice while a dedicated backhaul does not. Up and Down add or remove a relay.

Course Handouts

Built-in, zoomable copies of WLAN Pros published reference cards: the 2.4, 5 and 6 GHz channel allocations (including the 6 GHz card with GVP, Geofenced Variable Power), the MCS index, Troubleshooting Causes, the Bubble Diagram, and the four checklists. Put one on the projector while you talk through it, and students have the same card in their own copy of the app.

About the labs

The appendix has one lab per simulator. Each lab walks a student through the tool click by click, with a screenshot of every step and a close-up of what to click, so it works without an instructor in the room. Each exercise asks for a prediction first, then shows the answer on screen. Every lab comes in two versions: a student version without the answers, and an instructor version with them.

Use a lab after you have demonstrated the tool in class, as a small-group activity, or as homework. The lesson sequences below list the labs that go with each one.

Lesson sequences that work

These are starting points. Each takes about half an hour of demonstration and discussion.

- **Why 6 GHz behaves differently.** FSPL Simulator (the extra loss comes from the antenna, not the air), then Wi-Fi Through a Wall, then 6 GHz Power and PSD. Suggested labs: FSPL Simulator, Wi-Fi Through a Wall, 6 GHz Power and PSD.
- **RF math and the link budget.** FSPL Simulator, then the Antenna Fundamentals lesson and Antenna Pattern, then Rate vs Range, then Uplink vs Downlink. Suggested labs: FSPL Simulator, Antenna Pattern, Rate vs Range, Uplink vs Downlink.
- **Why the signal jumps when you move.** Multipath Simulator, then Room Propagation's close-up of the nulls near a wall. Suggested labs: Multipath Simulator, Room Propagation.
- **Why is the Wi-Fi slow?** Medium Access Simulator, then Airtime Anatomy, then Airtime Fairness, then Rate Adaptation. Suggested labs: Airtime Anatomy, Airtime Fairness.
- **Troubleshooting slow or unstable Wi-Fi.** Channel Utilization Meter, then Rate Adaptation, Airtime Fairness, Multicast at the Basic Rate and Legacy Protection Cost, then Roaming Walk and Why Two Devices Disagree. Suggested labs: Channel Utilization Meter, Multicast at the Basic Rate, Legacy Protection Cost, Why Two Devices Disagree.
- **QoS and contention.** Medium Access Simulator with EDCA priority, then Airtime Anatomy with a different access category, then Power Save for voice latency, then Channel Utilization Meter. Suggested labs: Medium Access Simulator, Airtime Anatomy, Power Save.
- **Antennas.** The Antenna Fundamentals lesson, then Antenna Pattern. Suggested lab: Antenna Pattern.
- **Spectrum analysis.** The Spectrum Analysis lesson, then the Swept vs FFT race and OFDM modes in Fourier and FFT. Suggested lab: Fourier and FFT.
- **From bits to rate.** Modulation Simulator, then Rate vs Range, then PHY Preamble Reference. Suggested labs: Modulation Simulator, Rate vs Range.

- **Channels and the rules.** The 2.4, 5 and 6 GHz channel cards in Course Handouts, then Channel Planner, DFS and Radar, 6 GHz Power and PSD, and Adjacent Channels and AP Stacking. Suggested labs: Channel Planner, DFS and Radar, Adjacent Channels and AP Stacking.
- **Designing a cell plan.** Channel Planner, then Roaming Walk, then DFS and Radar. Suggested labs: Channel Planner, Roaming Walk.
- **Wi-Fi 6 and 7 features.** OFDMA Resource Units, Spatial Reuse, then Multi-Link Operation. Suggested labs: OFDMA Resource Units, Spatial Reuse, Multi-Link Operation.
- **Association and roaming.** The scan, authentication, association and the 4-way handshake, one frame at a time. Association, Frame by Frame, then the 802.1X ladder in roam mode, then Roaming Walk. Suggested labs: Association, Frame by Frame; Roaming Walk.
- **Wi-Fi security modes.** Association, Frame by Frame for the 4-way handshake, then 802.1X and EAP Ladder run once for EAP-TLS and once for PEAP, then PSK and SAE for contrast. Suggested lab: 802.1X and EAP Ladder.
- **Why clients misbehave.** Why Two Devices Disagree, then Uplink vs Downlink, then Band Steering. Suggested labs: Why Two Devices Disagree, Band Steering.
- **Where the airtime goes.** Multicast at the Basic Rate, then Channel Utilization Meter, then Legacy Protection Cost. Suggested labs: Multicast at the Basic Rate, Legacy Protection Cost.
- **Surveys and heat maps.** Survey Walk, then Heat Map Builder, then Predict, Then Measure. Suggested labs: Survey Walk; Heat Map Builder; Predict, Then Measure.
- **Find My.** The Find My, Explained lesson, with Figure 3 (where the dot on the map comes from) as the Wi-Fi hook for a Wi-Fi audience.
- **Starlink.** The Starlink, Explained lesson, with Figure 9 (two networks in one house) as the Wi-Fi hook for a Wi-Fi audience.

What the numbers are, and what they are not

The simulators are built from the IEEE 802.11 standard, ITU recommendations and FCC and ETSI rules, and each tool's help says where its numbers come from. Some values are teaching models rather than measurements, and the screens say so where that is the case: a success curve that stands in for a real radio, a timing chosen to illustrate, the receiver sensitivities that are conformance floors real radios beat. When a student asks "is that what my AP does?", the answer is often "this is the mechanism; your AP's numbers will differ", and the tool's help tells you which parts are which.

After class

Students have the same app. Point them to the Wireless Classroom on a laptop or tablet, and suggest they repeat the demonstrations you did, changing one control at a time. The handouts give them the cards to keep, and the Field Manual in the app documents every tool in detail.

For homework, assign the student version of one lab from the day's sequence. Ask for the predictions they wrote down and one sentence on what surprised them.

Feedback and new lessons

We will be adding more guided lessons in the future. If there are other lessons you like to teach your students, please contact Keith@wlanpros.com and he'll work with you to get your ideas incorporated into a future update.